PROBLEM VF12

Journal Bearing

Original statement · VF12.pdf

Eccentric cylinders | Wedge effect | Force, torque, cavitation

OFFICIAL PROBLEM STATEMENT

The cylindrical bearing shown in the figure includes an external fixed casing of radius R_b and a shaft rotating with angular velocity ω of radius R such that $R_b - R \ll R$. The centers are shifted a small distance e, called exentricity, so that in between both cylinders there exists a thin circular gap of thickness $h \sim e \sim R_b - R \ll R$ full of a liquid of density ρ and viscosity μ.

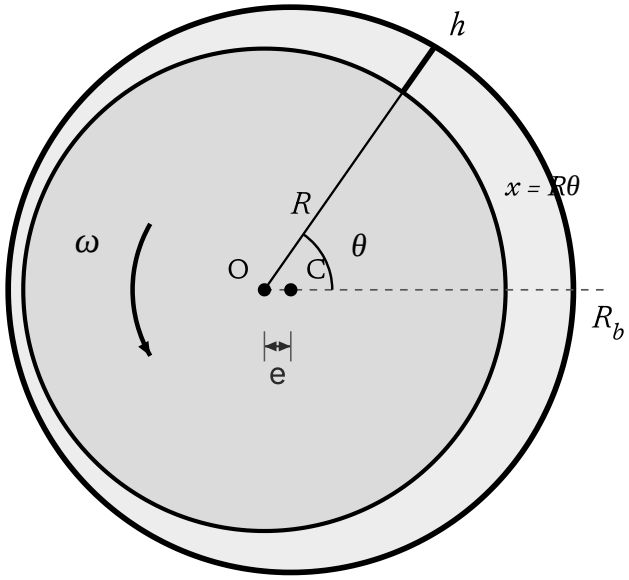
1. Show by simple trigonometric reasoning that the gap thickness can be expressed as $h(\theta) = R_b - R + e\cos\theta$.

2. Obtain the azimuthal velocity distribution v_x in terms of the distance y to the fixed casing and the unknown value of the induced pressure gradient $P_l = -R^{-1}\partial P/\partial\theta$.

3. Determine the circulating volume flux $Q = \int_0^h v_x\, dy$ as well as the pressure distribution $P(\theta) - P(0)$, where $P(0)$ is the unknown value of the pressure at $\theta = 0$, which depends on the pressurization conditions.

4. Calculate the torque and the force acting on the rotating shaft.

The sheet prints "exentricity". In the figure and in the key y is measured from the shaft surface (the key's profile is ωR at $y = 0$ and zero at $y = h$), not from the casing as the sentence of item 2 says.



Statement figure (redrawn from VF12.pdf): a shaft of radius R rotating with ω inside a fixed casing of radius R_b, their centres O and C a distance e apart; θ is measured from the line of centres, h(θ) is the gap, x = Rθ the arc along the shaft and y the distance across the gap, from the shaft to the casing

Source: VF12.pdf (page 1, statement and handwritten key; no date printed). Notes.pdf, Chapter 7: Couette and Poiseuille flows, Eqs. 7.9-7.12; the wedge effect, Eqs. 7.69-7.76.

Worked solution

Hypotheses, Data and Unknowns

- The gap is thin ($h \sim e \sim R_b - R \ll R$), so it can be unrolled: $x = R\theta$ along the shaft and y across the gap, with $y = 0$ on the shaft (moving with ωR) and $y = h$ on the fixed casing; the flow is steady and slender, with the pressure uniform across the gap and $P_l = -R^{-1}\partial P/\partial\theta$.
- Let $c = R_b - R$ and $\varepsilon = e/c$ ($0 \le \varepsilon < 1$). Forces are per unit length of the shaft.

Symbol	Meaning	SI unit
R, R_b	radii of the shaft and of the casing	m
e, $\varepsilon = e/(R_b - R)$	eccentricity and its relative value	m, 1
ω	angular velocity of the shaft	s^{-1}

Symbol	Meaning	SI unit
Q	circulating flux per unit length	$\mathrm{m^2/s}$

Gap Thickness

Place the centre of the casing at C and the centre of the shaft at O, a distance e from C, with θ the angle at O from the line OC direction; the radial line from O at the angle θ meets the casing at a point at the distance $R+h$ from O. In the triangle O, C, point of the casing, the sides are $R+h$ (from O), e (from C to O) and R_b (from C), with the angle θ at O between the sides $R+h$ and e; the law of cosines gives (the legs of the key's sketch, $(R+h)\cos\theta - e$ and $(R+h)\sin\theta$, give the same):

$$R_b^2 = (R+h)^2 - 2e(R+h)\cos\theta + e^2 \implies R+h = e\cos\theta + \sqrt{R_b^2 - e^2\sin^2\theta}$$

The square root is $R_b\left[1 - e^2\sin^2\theta/(2R_b^2) + \dots\right]$, and the second term is a factor $e/R_b \ll 1$ smaller than e, so $R+h \simeq R_b + e\cos\theta$:

$$h(\theta) = R_b - R + e\cos\theta = (R_b - R)\left[1 + \varepsilon\cos\theta\right]$$

The gap is largest at $\theta = 0$ and smallest at $\theta = \pi$.

Velocity Profile, Flux and Pressure

The shaft moves with ωR and the casing is at rest: Couette plus Poiseuille (Eq. 7.70), with the pressure gradient along $x = R\theta$:

$$v_x = \omega R\,\frac{h-y}{h} - \frac{y(h-y)}{2\mu}\,\frac{1}{R}\frac{\partial P}{\partial\theta}, \qquad Q = \int_0^h v_x\,dy = \frac{\omega R h}{2} - \frac{h^3}{12\mu R}\frac{\partial P}{\partial\theta}$$

The flux is the same at every θ (Eq. 7.72), so $\partial P/\partial\theta = \frac{12\mu R}{h^3}\left(\frac{\omega R h}{2} - Q\right)$, and integrating from 0 to θ (Eq. 7.74):

$$P(\theta) - P(0) = 6\mu\omega R^2 \int_0^\theta \frac{d\theta'}{h^2} - 12\mu R Q \int_0^\theta \frac{d\theta'}{h^3}$$

Periodicity. $P(2\pi) = P(0)$ gives $Q = \frac{\omega R}{2} \int_0^{2\pi} \frac{d\theta}{h^2} \Big/ \int_0^{2\pi} \frac{d\theta}{h^3}$. With $h = c(1+\varepsilon\cos\theta)$ and the integrals $J_n = \int_0^{2\pi} d\theta/(1+\varepsilon\cos\theta)^n$ (computed here, with $|\varepsilon| < 1$: the standard result $J_1(a) = \int_0^{2\pi} \frac{d\theta}{a+\varepsilon\cos\theta} = \frac{2\pi}{\sqrt{a^2-\varepsilon^2}}$ follows from the substitution $t = \tan(\theta/2)$, and differentiating with respect to a gives $J_2(a) = -J_1' = \frac{2\pi a}{(a^2-\varepsilon^2)^{3/2}}$ and $J_3(a) = -\frac{1}{2}J_2' = \frac{\pi(2a^2+\varepsilon^2)}{(a^2-\varepsilon^2)^{5/2}}$, evaluated at $a = 1$)

$$\int_0^{2\pi} \frac{d\theta}{(1+\varepsilon\cos\theta)^2} = \frac{2\pi}{(1-\varepsilon^2)^{3/2}}, \qquad \int_0^{2\pi} \frac{d\theta}{(1+\varepsilon\cos\theta)^3} = \frac{\pi\left(2+\varepsilon^2\right)}{(1-\varepsilon^2)^{5/2}}$$

$$Q = \frac{\omega R c}{2} \, \frac{2\pi/(1-\varepsilon^2)^{3/2}}{\pi(2+\varepsilon^2)/(1-\varepsilon^2)^{5/2}} = \omega R c \, \frac{1-\varepsilon^2}{2+\varepsilon^2}$$

Pressure. Substituting Q:

$$\frac{\partial P}{\partial\theta} = \frac{6\mu\omega R^2}{c^2}\left[\frac{1}{(1+\varepsilon\cos\theta)^2} - \frac{2(1-\varepsilon^2)}{(2+\varepsilon^2)(1+\varepsilon\cos\theta)^3}\right] = \frac{6\mu\omega R^2}{c^2(2+\varepsilon^2)} \, \frac{\varepsilon\left(2\cos\theta + 3\varepsilon + \varepsilon^2\cos\theta\right)}{(1+\varepsilon\cos\theta)^3}$$

and this is the derivative of

$$P(\theta) - P(0) = \frac{6\mu\omega R^2}{(R_b - R)^2} \, \frac{\varepsilon\,(2+\varepsilon\cos\theta)\sin\theta}{(2+\varepsilon^2)(1+\varepsilon\cos\theta)^2}$$

(check: with $N = (2+\varepsilon\cos\theta)\sin\theta$, $N' = 2\cos\theta - \varepsilon + 2\varepsilon\cos^2\theta$ and $\frac{d}{d\theta}\frac{N}{(1+\varepsilon\cos\theta)^2} = \frac{N'(1+\varepsilon\cos\theta)+2\varepsilon\sin\theta\, N}{(1+\varepsilon\cos\theta)^3} = \frac{2\cos\theta+3\varepsilon+\varepsilon^2\cos\theta}{(1+\varepsilon\cos\theta)^3}$, which has the factor ε of the previous line; and the expression vanishes at $\theta = 0$). The pressure is antisymmetric, $P(\theta) - P(0) = -[P(-\theta) -$

$P(0)]$: overpressure on one side of the line of centres and underpressure on the other, which gives a force perpendicular to it.

Force and Torque on the Shaft

Torque. The shear on the shaft (normal from the shaft into the liquid, $+\bar{e}_y$) is $\tau_p = \mu\,\partial v_x/\partial y|_{y=0} = -\frac{\mu\omega R}{h} - \frac{h}{2R}\frac{\partial P}{\partial\theta}$; with $\partial P/\partial\theta = 6\mu\omega R^2/h^2 - 12\mu RQ/h^3$, $\tau_p = -\frac{4\mu\omega R}{h} + \frac{6\mu Q}{h^2}$. The torque per unit length is $M = \int_0^{2\pi} \tau_p R\cdot R\,d\theta$:

$$M = -4\mu\omega R^3\int_0^{2\pi}\frac{d\theta}{h} + 6\mu Q R^2\int_0^{2\pi}\frac{d\theta}{h^2}, \qquad \int_0^{2\pi}\frac{d\theta}{1+\varepsilon\cos\theta} = \frac{2\pi}{\sqrt{1-\varepsilon^2}}$$

$$M = -\frac{8\pi\mu\omega R^3}{c\sqrt{1-\varepsilon^2}} + \frac{12\pi\mu\omega R^3}{c(2+\varepsilon^2)\sqrt{1-\varepsilon^2}} = -\frac{4\pi\mu\omega R^3}{R_b - R}\,\frac{1+2\varepsilon^2}{\sqrt{1-\varepsilon^2}\,(2+\varepsilon^2)}$$

(the torque of the liquid on the shaft, opposing the rotation; the torque the motor applies is its opposite).

Force. The pressure force on the shaft is $\bar{F} = -\int (P - P(0))\,\bar{n}\,R\,d\theta$, with $\bar{n}$ the outward normal of the shaft, $n_z = \sin\theta$ and $n_{\parallel} = \cos\theta$ (the uniform $P(0)$ gives no net force). The component along the line of centres is zero because $P - P(0)$ is odd in θ and $\cos\theta$ is even. The perpendicular one, integrating by parts ($P(0) = P(2\pi)$ so the boundary term vanishes): $\int_0^{2\pi}(P-P(0))\sin\theta\,d\theta = \int_0^{2\pi}\cos\theta\,\frac{\partial P}{\partial\theta}\,d\theta = \frac{6\mu\omega R^2\varepsilon}{c^2(2+\varepsilon^2)}\int_0^{2\pi}\frac{\cos\theta\left(2\cos\theta+3\varepsilon+\varepsilon^2\cos\theta\right)}{(1+\varepsilon\cos\theta)^3}d\theta$, and the last integral equals $2\pi/\sqrt{1-\varepsilon^2}$. To see it, write $u = 1+\varepsilon\cos\theta$, $\cos\theta = (u-1)/\varepsilon$: the numerator is $\cos\theta\left[(2+\varepsilon^2)\cos\theta + 3\varepsilon\right] = \frac{2+\varepsilon^2}{\varepsilon^2}(u-1)^2 + 3(u-1)$, so the integral is $\frac{2+\varepsilon^2}{\varepsilon^2}\,(J_1 - 2J_2 + J_3) + 3\,(J_2 - J_3)$. With $s = \sqrt{1-\varepsilon^2}$, $J_1 - 2J_2 + J_3 = \frac{\pi}{s^5}\left[2s^4 - 4s^2 + 2 + \varepsilon^2\right] = \frac{\pi\varepsilon^2(1+2\varepsilon^2)}{s^5}$ and $J_2 - J_3 = \frac{\pi}{s^5}\left[2s^2 - 2 - \varepsilon^2\right] = -\frac{3\pi\varepsilon^2}{s^5}$, and the sum is $\frac{\pi}{s^5}\left[(2+\varepsilon^2)(1+2\varepsilon^2) - 9\varepsilon^2\right] = \frac{2\pi(1-\varepsilon^2)^2}{s^5} = \frac{2\pi}{s}$ (also confirmed numerically by `verify_viscous_flows.py`). Hence

$$F_z = -R\int_0^{2\pi}(P-P(0))\sin\theta\,d\theta = -\frac{12\pi\mu\omega R^3}{(R_b-R)^2}\,\frac{\varepsilon}{\sqrt{1-\varepsilon^2}\,(2+\varepsilon^2)}$$

on the shaft (downward in the figure), and the opposite on the bearing (upward). The shear forces contribute to the force on the shaft at a relative order $c/R \ll 1$ and are neglected.

Limit of Large Eccentricity and Cavitation

When $\varepsilon \to 1$ (the shaft touches the casing, $h_{min} = c(1-\varepsilon) \to 0$) the force grows without bound, $F_z \propto (1-\varepsilon^2)^{-1/2}$: **a bearing can in principle bear an infinite load**. In practice it fails earlier: the underpressure side of $P(\theta) - P(0)$ reaches the vapour pressure p_v and the liquid cavitates when $P_{min} - P(0) = p_v - P(0)$. The minimum of P is where $\partial P/\partial\theta = 0$, that is $\cos\theta_{min} = -3\varepsilon/(2+\varepsilon^2)$ with $\sin\theta_{min} < 0$ ($\theta_{min} \simeq 3\pi/2$ for small ε, as the key notes).

Answers and Checks

- Gap: $h = R_b - R + e\cos\theta = (R_b - R)(1 + \varepsilon\cos\theta)$.
- Velocity: $v_x = \omega R\frac{h-y}{h} - \frac{y(h-y)}{2\mu R}\frac{\partial P}{\partial\theta}$; flux: $Q = \omega R(R_b - R)\frac{1-\varepsilon^2}{2+\varepsilon^2}$.
- Pressure: $P - P(0) = \frac{6\mu\omega R^2}{(R_b-R)^2}\frac{\varepsilon(2+\varepsilon\cos\theta)\sin\theta}{(2+\varepsilon^2)(1+\varepsilon\cos\theta)^2}$.
- Torque: $M = -\frac{4\pi\mu\omega R^3}{R_b-R}\frac{1+2\varepsilon^2}{\sqrt{1-\varepsilon^2}(2+\varepsilon^2)}$; force: $F_z = -\frac{12\pi\mu\omega R^3}{(R_b-R)^2}\frac{\varepsilon}{\sqrt{1-\varepsilon^2}(2+\varepsilon^2)}$ on the shaft, opposite on the bearing.
- Limit $\varepsilon \to 0$ (concentric): $Q = \omega Rc/2$, $P = P(0)$, $F_z = 0$ and $M = -2\pi\mu\omega R^3/c$, the Couette torque of the gap c: $\tau = \mu\omega R/c$ times $R \cdot R \cdot 2\pi$.
- Dimensions: $\mu\omega R^3/(R_b - R)$ is $ML^{-1}T^{-1}\cdot T^{-1}\cdot L^3/L = MLT^{-2}$, a torque per unit length (N m/m = N); $\mu\omega R^3/(R_b-R)^2$ is a force per unit length.
- Check against the official key: VF12.pdf gives the same geometry, profile, flux, pressure, torque and force. Its part 1 writes the gap as $(R_b - R)[1 + \varepsilon\cos\theta]$ and its flux integrals as $(1 - \varepsilon\cos\theta)$; the two give the same integrals by $\theta \to \theta + \pi$.